CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

TASK 1 – Patient Data Representation and Normalization

a) Effect of using raw features without normalization

Patient data contains features such as age, blood pressure, glucose level and cholesterol. These features have different numerical scales.

For example:

- Age: 25–70
- Blood Pressure: 80–180
- Glucose: 70–250
- Cholesterol: 100–300

If raw features are used without normalization, features with larger numerical values may dominate the learning process.

Effects:

1. Features with larger scales may have more influence on the model.
2. Distance-based algorithms may calculate misleading distances.
3. Gradient-based algorithms may take inefficient steps during training.
4. Model convergence may become slower.

Therefore, normalization is necessary to bring features into a common scale.

b) Feature matrix and normalization

Consider three patients with the following data.

Original Feature Matrix:

Patient 1: Age = 20, BP = 100, Glucose = 100, Cholesterol = 150

Patient 2: Age = 40, BP = 120, Glucose = 150, Cholesterol = 200

Patient 3: Age = 60, BP = 140, Glucose = 200, Cholesterol = 250

Feature Matrix:

X =

[20 100 100 150]

[40 120 150 200]

[60 140 200 250]

Using Min-Max Normalization:

Formula:

X_normalized = (X - X_min) / (X_max - X_min)

For Age:

Minimum = 20, Maximum = 60

Patient 1:

(20 - 20) / (60 - 20) = 0

Patient 2:

(40 - 20) / (60 - 20) = 0.5

Patient 3:

(60 - 20) / (60 - 20) = 1

Similarly, we normalize all other features.

Normalized Feature Matrix:

[0.0 0.0 0.0 0.0]

[0.5 0.5 0.5 0.5]

[1.0 1.0 1.0 1.0]

c) Importance of normalization

1. It brings all features into a common numerical range.
2. It prevents large-scale features from dominating smaller-scale features.
3. It improves the performance of distance-based algorithms such as KNN and K-means.
4. It improves the convergence of gradient-based algorithms such as Gradient Descent.
5. It makes model training more stable and efficient.

Conclusion: Normalization improves the reliability and efficiency of machine learning models.

TASK 2 – Vector Representation and Matrix Operations

a) Vector representation

A patient's features can be represented as a vector.

Consider one patient:

Age = 40

Glucose = 150

Blood Pressure = 120

Cholesterol = 200

Patient Feature Vector:

X = [40, 150, 120, 200]

Importance Weight Vector:

W = [0.2, 0.4, 0.2, 0.2]

Weighted score is calculated using the dot product.

Formula:

Risk Score = X . W

Risk Score = (Age × Weight) + (Glucose × Weight) + (BP × Weight) + (Cholesterol × Weight)

b) Numerical calculation

Risk Score = (40 × 0.2) + (150 × 0.4) + (120 × 0.2) + (200 × 0.2)

= 8 + 60 + 24 + 40

= 132

Final Risk Score = 132

Interpretation:

The weighted score is 132. Glucose has the highest importance weight of 0.4, so it contributes more to the preliminary score.

Conclusion: Vector and matrix operations help calculate patient risk scores efficiently.

TASK 3 – Probability and Conditional Probability Given:

Total patients = 1000

Patients with disease = 200

Patients without disease = 800

Diseased patients with elevated glucose = 160

Non-diseased patients with elevated glucose = 120

Let:

D = Patient has disease.

G = Patient has elevated glucose.

a) Probability of elevated glucose

Total patients with elevated glucose:

= 160 + 120

= 280

Formula:

P(G) = Number of patients with elevated glucose / Total patients

P(G) = 280 / 1000

P(G) = 0.28 = 28%

b) Probability of disease given elevated glucose

Formula:

$P(D|G) = P(D \cap G) / P(G)$

Substituting the values:

P(D|G) = 160 / 280

P(D|G) = 0.5714

P(D|G) = 57.14%

Interpretation:

There is a 57.14% probability that a patient has the disease when the patient has elevated glucose.

c) Probability of elevated glucose given disease

Formula:

P(G|D) = P(G ∩ D) / P(D)

Substituting the values:

P(G|D) = 160 / 200

P(G|D) = 0.8

P(G|D) = 80%

Interpretation:

80% of patients who have the disease also have elevated glucose.

d) Importance of conditional probability in healthcare

1. It helps identify patients who may have a higher risk of disease.
2. It supports doctors in prioritizing patients for further medical testing.
3. It helps understand the relationship between symptoms and diseases.
4. It supports better healthcare decision-making.

Conclusion: Conditional probability helps hospitals estimate disease risk using patient information.

TASK 4 – Bayesian Statistics for Disease Classification

Given:

Probability of disease = 10% = 0.10

Probability of no disease = 90% = 0.90

Probability of positive test given disease = 90% = 0.90

Probability of positive test given no disease = 5% = 0.05

a) Calculate probability of disease given a positive test

Using Bayes' Theorem:

Formula:

$P(D|T) = [P(T|D) \times P(D)] / P(T)$

Where:

P(D) = Prior probability

P(T|D) = Likelihood

P(D|T) = Posterior probability

First, calculate the probability of a positive test.

$P(T) = P(T|D) \times P(D) + P(T|\neg D) \times P(\neg D)$

Substituting:

$P(T) = (0.90 \times 0.10) + (0.05 \times 0.90)$

$P(T) = 0.09 + 0.045$

$P(T) = 0.135$

Now apply Bayes' theorem:

$P(D|T) = (0.90 \times 0.10) / 0.135$

$P(D|T) = 0.09 / 0.135$

$P(D|T) = 0.6667$

Final Answer = 66.67%

b) Roles of prior probability and likelihood

1. Prior Probability: It represents the initial probability of a patient having the disease before conducting the test. Here, it is 10%.

2. Likelihood: It represents the probability of obtaining a positive test result when the patient actually has the disease. Here, it is 90%.
3. Posterior Probability: It represents the updated probability of disease after considering the test result.

c) Importance of Bayesian reasoning

1. It helps make decisions when medical information is uncertain.
2. It combines previous knowledge with new test results.
3. It updates disease probability when new evidence becomes available.
4. It supports medical diagnosis and risk prediction.

Conclusion: Bayesian statistics helps doctors make better decisions by updating disease probabilities using available evidence.

TASK 5 – Entropy and Information Gain

Given:

Total records = 100

High-risk patients = 40

Low-risk patients = 60

Probability of high risk:

P(High) = 40/100 = 0.4

Probability of low risk:

P(Low) = 60/100 = 0.6

a) Calculate entropy

Formula:

Entropy(H) = $-\Sigma P(x) \log_2 P(x)$

Substituting:

$H = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)]$

$H = -[(0.4 \times -1.3219) + (0.6 \times -0.7370)]$

H = -[-0.5288 - 0.4422]

Entropy = 0.9710 bits

b) Meaning of entropy

Entropy measures the uncertainty or impurity in a dataset.

1. High entropy indicates greater uncertainty.
2. Low entropy indicates less uncertainty.
3. Zero entropy means all records belong to the same class.

Here, entropy is 0.9710 bits, indicating relatively high uncertainty because the dataset contains both high-risk and low-risk patients.

c) Information gain

Information Gain measures how much uncertainty is reduced after splitting the dataset using a feature.

Formula:

Information Gain = Entropy(Parent) - Weighted Entropy(Children)

For example, glucose level can divide patients into groups based on normal and elevated glucose.

If the split produces groups with fewer mixed risk labels, entropy decreases and information gain increases.

A feature with higher information gain is generally more useful for classification.

Conclusion: Entropy measures uncertainty, while information gain helps select important features for disease prediction.

TASK 6 – Numerical Computation and Distance Measurement

Given:

Patient A = (0.70, 0.80, 0.60)

Patient B = (0.50, 0.75, 0.90)

a) Calculate Euclidean distance

Formula:

$D(A,B) = \sqrt{[(x1-y1)^2 + (x2-y2)^2 + (x3-y3)^2]}$

Substituting:

$D(A,B) = \sqrt{[(0.70-0.50)^2 + (0.80-0.75)^2 + (0.60-0.90)^2]}$

$= \sqrt{[(0.20)^2 + (0.05)^2 + (-0.30)^2]}$

$= \sqrt{[0.04 + 0.0025 + 0.09]}$

$= \sqrt{0.1325}$

$D(A,B) = 0.3640$

b) Interpretation of distance

The Euclidean distance between the two patients is 0.3640.

A smaller distance indicates that the patients have more similar feature values.

A larger distance indicates that their characteristics are more different.

This measurement can help identify similar patients in healthcare datasets.

c) Importance of normalization

1. Features have different numerical scales.
2. Large-scale features may dominate distance calculations.
3. Normalization gives equal consideration to features on a common scale.
4. It improves the reliability of KNN and K-means algorithms.

Conclusion: Euclidean distance measures patient similarity, while normalization ensures fair comparison between features.

TASK 7 – Bias–Variance Trade-off

Given:

Model A: High training error and high validation error.

Model B: Very low training error and high validation error.

Model C: Low training error and low validation error.

a) Identify bias and variance

Model A – High Bias (Underfitting)

Model B – High Variance (Overfitting)

Model C – Good Generalization

b) Effect of model complexity on bias and variance

Bias:

Bias is the error caused by overly simple assumptions in a learning algorithm.

High bias occurs when a model is too simple to learn important patterns.

Variance:

Variance is the model's sensitivity to changes in training data.

High variance occurs when a model learns training data too closely, including noise.

Effect of complexity:

1. Low model complexity generally produces high bias and low variance.
2. Increasing complexity reduces bias but may increase variance.
3. Very high complexity may cause overfitting.
4. A suitable model balances bias and variance.

c) Improvements for Models A and B

Model A – High Bias:

1. Increase model complexity.
2. Add useful features.
3. Reduce excessive regularization.
4. Use a more flexible learning algorithm.

Model B – High Variance:

1. Apply regularization.
2. Increase training data.
3. Reduce model complexity.
4. Use cross-validation.

5. Apply feature selection.

Conclusion: A good machine learning model should maintain a balance between bias and variance to achieve better performance on unseen patient records.

TASK 8 – Overfitting, Underfitting and Model Capacity

a) Identify the model behaviour

1. Simple model that fails to capture disease patterns – Underfitting.
2. Moderately complex model that performs consistently – Good Generalization.
3. Highly complex model with excellent training performance but poor unseen performance – Overfitting.

b) Model capacity

Model capacity refers to the ability of a machine learning model to learn different patterns and relationships from data.

1. Low capacity models learn only simple relationships.
2. Medium capacity models learn useful patterns and generalize well.
3. High capacity models can learn complex relationships but may also memorize noise.

c) Why increasing capacity first improves performance but later causes overfitting

When model capacity increases:

1. Initially, the model learns more important patterns.
2. Training and testing performance may improve.
3. If capacity becomes too high, the model may memorize training examples.
4. It may learn noise instead of general patterns.
5. As a result, performance on unseen data decreases.

Conclusion: Selecting suitable model capacity is important for achieving good generalization and avoiding overfitting and underfitting.

TASK 9 – Supervised Learning and Model Evaluation (3 Marks)

a) Why is this a supervised learning problem?

The hospital has historical patient records with known disease labels.

The labels are:

- Diseased
- Disease-free

Supervised learning uses labelled training data to learn the relationship between patient features and disease outcomes.

The trained model can predict disease status for new patients.

Therefore, this is a supervised learning problem.

b) Roles of training and testing sets

Training Set:

1. Used to train the machine learning model.
2. Helps the model learn patterns from patient records.
3. Used to estimate model parameters.

Testing Set:

1. Used to evaluate the trained model.
2. Contains records not used during training.
3. Measures how well the model performs on unseen patients.

c) Calculate accuracy

Given:

Total test patients = 100

Correctly classified patients = 92

Formula:

Accuracy = (Correct Predictions / Total Predictions) × 100

Accuracy = (92/100) × 100

Accuracy = 92%

Interpretation:

The model correctly predicts the disease status of 92 out of 100 patients.

Conclusion: Accuracy measures the overall correctness of a classification model. For healthcare, sensitivity and specificity should also be considered.

TASK 10 – Unsupervised Learning and Patient Grouping

a) Why is unsupervised learning suitable?

The hospital has patient records without predefined disease labels.

Unsupervised learning is suitable because it discovers hidden patterns and groups in unlabelled data.

It helps identify patients with similar health characteristics without requiring predefined categories.

b) How K-means groups patients

K-means is an unsupervised clustering algorithm that divides data into K groups.

Steps:

1. Select the number of clusters K.
2. Initialize K cluster centroids.
3. Calculate the distance between each patient and the centroids.
4. Assign each patient to the nearest centroid.
5. Recalculate the centroid of each cluster.
6. Repeat until the cluster assignments become stable.

Example:

K = 3

Cluster 1 – Patients with relatively normal health measurements.

Cluster 2 – Patients with moderately elevated glucose and blood pressure.

Cluster 3 – Patients with high glucose and cholesterol levels.

c) Use of distance to measure similarity

K-means uses Euclidean distance to measure similarity between patients.

Formula:

$D = \sqrt{[(x1-y1)^2 + (x2-y2)^2 + ... + (xn-yn)^2]}$

Patients with smaller distances are considered more similar.

Patients are assigned to the nearest cluster centroid.

d) Clinical importance of clusters

1. Helps identify groups of patients with similar health characteristics.
2. Supports early identification of unusual patient patterns.
3. Helps doctors prioritize patients for further investigation.
4. Supports healthcare planning and resource allocation.

Clustering results should support clinical investigation and must not replace professional medical diagnosis.

Conclusion: K-means helps hospitals discover hidden patient groups from unlabelled healthcare data.

TASK 11 – Comparison of Supervised and Unsupervised Learning

a) Identify suitable learning paradigms

Dataset A:

Contains patient features with disease labels.

Suitable approach: Supervised Learning.

Reason: The model learns from labelled examples and predicts disease status.

Dataset B:

Contains patient features without disease labels.

Suitable approach: Unsupervised Learning.

Reason: The model discovers hidden patterns and groups patients based on similarities.

b) Compare objectives and outputs

Supervised Learning:

1. Uses labelled data.
2. Learns the relationship between input features and target labels.
3. Predicts outcomes for new records.

4. Example: Disease classification.
5. Output: Predicted disease or disease-free class.

Unsupervised Learning:

1. Uses unlabelled data.
2. Identifies hidden patterns and similarities.
3. Groups similar records.
4. Example: K-means clustering.
5. Output: Patient clusters or groups.

c) Recommend an approach for predicting disease status

Supervised learning is recommended for predicting disease status in a new patient.

Reason:

1. Historical records contain known disease labels.
2. The model can learn from previous patient outcomes.
3. It can predict whether a new patient is likely to have the disease.
4. Its performance can be evaluated using accuracy, precision, recall and F1-score.

Conclusion: Supervised learning is suitable for disease prediction, while unsupervised learning is suitable for discovering hidden patient groups. Both approaches can be used together for better healthcare analysis.

TASK 12 – DESIGN CHALLENGE: SELECTING AND ADAPTING A LEARNING STRATEGY

Changed Requirement:

The hospital initially analyzes 10,000 patient records. Now, it must process 1,000,000 records, predict disease risk, discover patient groups and handle missing values and different feature scales.

The analytical strategy should be modified by combining supervised and unsupervised learning.

A. Impact Analysis (2 Marks)

a) Effect of increasing dataset size

Increasing the dataset from 10,000 to 1,000,000 records affects the learning system in several ways.

1. Data Representation:

The feature matrix becomes larger, requiring more memory and storage.

For example:

10,000 records × 4 features = 40,000 feature values.

1,000,000 records × 4 features = 4,000,000 feature values.

2. Numerical Computation:

More records increase the computational operations required for training and prediction.

3. Learning Efficiency:

Training may take more time and require efficient algorithms, batch processing and suitable hardware.

4. Model Performance:

A larger and more diverse dataset may improve generalization if the data is representative and of good quality.

b) Importance of normalization, missing-value handling and data variability

Normalization:

Brings different features into a common scale and improves distance calculations.

Missing-Value Handling:

Missing values can affect model training and reduce prediction accuracy.

They can be handled using mean, median or mode imputation, depending on the feature and data distribution.

Data Variability:

Different patient characteristics help the model learn more general patterns.

A diverse dataset reduces the risk of learning patterns that apply only to a small group of patients.

Conclusion: Proper data preprocessing is necessary to maintain data quality and improve learning efficiency.

B. Evaluation of the Original Approach

a) Is a single supervised approach sufficient?

No. A single supervised learning approach is not sufficient for both disease prediction and patient-group discovery.

Supervised learning:

- Uses labelled patient records.
- Predicts disease risk.
- Classifies patients into disease categories.

Unsupervised learning:

- Uses unlabelled patient records.
- Discovers hidden patterns.
- Groups patients with similar characteristics.

Therefore, both methods are required to satisfy the hospital's requirements.

b) Limitations of labelled learning

1. Supervised learning requires labelled training data.
2. Labelling large healthcare datasets is expensive and time-consuming.
3. Unlabelled records cannot directly contribute as labelled examples in ordinary supervised training.
4. Hidden patterns in unlabelled data may remain undiscovered.
5. The model may not capture all patient groups if only labelled data is used.

Conclusion: Combining supervised and unsupervised learning helps use both labelled and unlabelled patient records effectively.

C. Adaptation or Redesign

a) Proposed strategy combining supervised and unsupervised learning

The hospital should use a hybrid machine learning approach.

Proposed steps:

Step 1: Data Collection

Collect 1,000,000 patient records containing age, blood pressure, glucose and cholesterol.

Step 2: Data Preprocessing

- Handle missing values using suitable imputation.
- Remove duplicate records.
- Normalize numerical features.
- Check for incorrect or inconsistent values.

Step 3: Supervised Learning

Use a classification algorithm such as Random Forest or Logistic Regression.

Train the model using labelled patient records.

Purpose: Predict disease risk for new patients.

Step 4: Unsupervised Learning

Apply K-means clustering to patient records, including unlabelled records.

Purpose: Discover patient groups with similar health characteristics.

Step 5: Evaluation

Evaluate classification performance and clustering quality separately.

b) Incorporating mathematical concepts

1. Probability:

Use probability to estimate the likelihood of disease and support risk prediction.

2. Distance Measures:

Use Euclidean distance to identify similarities between patient records.

3. Normalization:

Apply Min-Max normalization to bring features into a common scale.

Formula:

$X' = (X - Xmin) / (Xmax - Xmin)$

4. Model Capacity:

Select suitable model complexity to avoid underfitting and overfitting.

5. Missing Values:

Use statistical imputation methods to handle incomplete patient records.

Conclusion: A hybrid strategy combines prediction and patient-group discovery while improving the use of available healthcare data.

D. Evidence-Based Defence

a) Proposed experiments and evaluation measures

The redesigned approach should be compared with the original supervised learning approach.

Experiment 1: Classification Performance

Train Logistic Regression and Random Forest using labelled records.

Evaluation measures:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Experiment 2: Clustering Performance

Apply K-means to patient records.

Evaluation measures:

- Silhouette Score
- Within-Cluster Sum of Squares (WCSS)
- Cluster Size

Experiment 3: Data Preprocessing

Compare model performance before and after normalization and missing-value handling.

Experiment 4: Computational Efficiency

Measure:

- Training time
- Prediction time
- Memory usage

b) Justification of the final design

The hybrid learning strategy is suitable because:

1. Supervised learning predicts disease risk using labelled patient records.
2. Unsupervised learning discovers hidden patient groups using unlabelled records.
3. Normalization ensures that all numerical features contribute fairly.
4. Missing-value handling improves data quality.
5. Probability supports reliable disease-risk estimation.
6. Suitable model capacity improves generalization and reduces overfitting.
7. Efficient algorithms and batch processing help manage 1,000,000 records.

The hospital should modify its original analytical strategy and adopt a hybrid machine learning approach combining supervised classification and unsupervised clustering.

This approach uses both labelled and unlabelled patient records, supports disease-risk prediction, discovers hidden patient groups and handles large-scale healthcare data efficiently.

The redesigned system should be evaluated using classification metrics, clustering measures and computational performance before clinical use.